

1. 求下列函数在有限奇点处的留数

(1) $f(z) = \frac{1+z}{z^2-2z}$

$z=0$ 和 $z=2$ 是 $f(z)$ 的一级极点.

$$\therefore \text{Res}[f(z), 0] = \lim_{z \rightarrow 0} (z-0) \cdot \frac{1+z}{z(z-2)} = -\frac{1}{2}$$

$$\text{Res}[f(z), z_0] = \lim_{z \rightarrow z_0} (z-z_0)f(z)$$

$$\text{Res}[f(z), 2] = \lim_{z \rightarrow 2} (z-2) \cdot \frac{1+z}{z(z-2)} = \frac{3}{2}$$

在 m 级极点处 $\text{Res}[f(z), z_0] = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z-z_0)^m f(z)]$

(2) $f(z) = \frac{1-e^{2z}}{z^3}$

$$f(z) = \frac{1}{z^3} \left[1 - \left(1 + 2z + \frac{1}{2!} \cdot (2z)^2 + \frac{1}{3!} \cdot (2z)^3 + \frac{1}{4!} \cdot (2z)^4 + \dots \right) \right]$$

$\therefore z=0$ 是 $f(z)$ 的三级极点.

$$\therefore \text{Res}[f(z), 0] = \frac{1}{(3-1)!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left(\frac{1-e^{2z}}{z} \right) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left[1 - (1 + 2z + 2z^2 + \frac{4}{3}z^3 + \dots) \right] = -\frac{4}{3}$$

(3) $f(z) = \cos \frac{1}{1-z}$

Laurent 级数无穷多负幂项. $\lim_{z \rightarrow z_0} f(z)$ 不存在且不为 ∞

$z=1$ 为本性奇点.

$\therefore$ 当 $0 < |z-1| < +\infty$ 时,

$$\cos \frac{1}{1-z} = 1 - \frac{1}{2!} \frac{1}{(1-z)^2} + \frac{1}{4!} \cdot \frac{1}{(1-z)^4} \dots + (-1)^n \cdot \frac{1}{(2n)!} \frac{1}{(1-z)^{2n}} \dots$$

$$\therefore C_{-1} = 0$$

留数: $C_{-1} = \frac{1}{2\pi i} \oint_C f(z) dz$ 用 Laurent 展开式

(4) $f(z) = \frac{1}{z \sin z}$

$z=0$ 为二级极点. $z=k\pi$ 为一级极点, $k=\pm 1, \pm 2, \dots$

$$\therefore \text{Res}[f(z), 0] = \frac{1}{(2-1)!} \lim_{z \rightarrow 0} \frac{d}{dz} \left(z^2 \cdot \frac{1}{z \sin z} \right) = 0$$

$$\text{Res}[f(z), k\pi] = \lim_{z \rightarrow k\pi} (z-k\pi) \cdot \frac{1}{z \sin z} = \lim_{z \rightarrow k\pi} \frac{z-k\pi}{z(z-k\pi) \cdot (-1)^k} = \frac{1}{k\pi} \cdot (-1)^k$$

2. 求 $\text{Res}[f(z), \infty]$ 的值

(1) $f(z) = \frac{e^z}{z^2-1}$

解: $f(z)$ 在扩充复平面上有奇点: $\pm 1, \infty$. ± 1 为 $f(z)$ 的一级极点.

$$\text{Res}[f(z), 1] = \lim_{z \rightarrow 1} (z-1) \cdot \frac{e^z}{(z+1)(z-1)} = \frac{1}{2}e$$

$$\text{Res}[f(z), -1] = \lim_{z \rightarrow -1} (z+1) \cdot \frac{e^z}{(z+1)(z-1)} = -\frac{1}{2}e^{-1}$$

$$\therefore \sum_{k=1}^{\infty} \text{Res}[f(z), z_k] + \text{Res}[f(z), \infty] = 0$$

$$\therefore \text{Res}[f(z), \infty] = -\text{sh } 1$$

求起导很烦

(2) $f(z) = \frac{1}{z(z+1)^2(z-4)}$

$$\text{Res}[f(z), \infty] = -\text{Res}\left[f\left(\frac{1}{z}\right) \cdot \frac{1}{z^2}, 0\right]$$

$$f\left(\frac{1}{z}\right) \cdot \frac{1}{z^2} = \frac{z^4}{(z+1)^2(z-4)}$$

$z=0$ 为可去奇点. 0 本性奇点 C_{-1}

$$\therefore \text{Res}\left[f\left(\frac{1}{z}\right) \cdot \frac{1}{z^2}, 0\right] = 0$$

$$\therefore \text{Res}[f(z), \infty] = 0.$$

3. 求下列积分值

函数在无穷远点的性质

无穷远点留数 $\rightarrow$ 零留数

(1) $\oint_C \frac{z^{15}}{(z^2+1)^2(z^2+2)^3} dz$, $C: |z|=3$

奇点有 $\infty, \pm i, \sqrt{2} e^{\frac{\pi+2k\pi}{4}i}, k=0,1,2,3$

只有 ∞ 不在 C 里面

$\therefore \oint_C f(z) dz = -2\pi i \operatorname{Res}[f(z), \infty] = 2\pi i \operatorname{Res}[f(\frac{1}{z}), \frac{1}{z}, 0]$

$\operatorname{Res}[f(z), \infty] = \frac{1}{2\pi i} \oint_C f(z) dz$

$f(z) \cdot \frac{1}{z^2} = \frac{1}{(z^2+1)^2(z^2+1)^3} \cdot \frac{1}{z}$

$= -\operatorname{Res}[f(\frac{1}{z}), \frac{1}{z}, 0]$

$z=0$ 为 -1 级极点

$\therefore \operatorname{Res}[f(\frac{1}{z}), \frac{1}{z}, 0] = \lim_{z \rightarrow 0} z \cdot f(z) \cdot \frac{1}{z^2} = 1$

$\therefore \oint_C f(z) dz = 2\pi i$

(3) $\int_0^{2\pi} \frac{1}{5+3\sin\theta} d\theta$

令 $z = e^{i\theta}$, $d\theta = \frac{dz}{zi}$, $0 \leq \theta \leq 2\pi$, $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

Cauchy 积分公式

$\oint_C \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0)$

$\therefore$ 原式 $= \oint_{|z|=1} \frac{2dz}{3z^2+10zi-3} = \frac{2}{3} \oint_{|z|=1} \frac{dz}{(z+3i)(z+\frac{1}{3})} = \frac{2}{3} \cdot \frac{1}{z+\frac{1}{3}} \Big|_{z=-\frac{1}{3}} \cdot 2\pi i = \frac{\pi}{2}$

(2) $\oint_C \frac{z^{2n}}{z^n+1} dz$ ($n \in \mathbb{Z}_+$) $C: |z|=r>1$

奇点为 ∞ , $z = e^{i\frac{\pi+2k\pi}{n}}$, $k=0,1,2,\dots,n-1$

$\therefore \oint_C f(z) dz = -2\pi i \operatorname{Res}[f(z), \infty]$

$\frac{z^{2n}}{1+z^n} = z^{2n} \sum_{k=0}^{\infty} (-1)^k \cdot \frac{1}{z^{kn}} = \sum_{k=0}^{\infty} (-1)^k \frac{1}{z^{(k-2)n}}$

$(k-2)n=1, n \geq 1, k \geq 0.$

$\therefore \operatorname{Res}[f(z), \infty] = -C_{-1} = \begin{cases} +1, & n=1 \\ 0, & n \geq 2 \end{cases}$

$\textcircled{1} n=1, k=3$

$\therefore \oint_C f(z) dz = \begin{cases} -2\pi i, & n=1 \\ 0, & n \geq 2. \end{cases}$

$\frac{1}{2\pi i} \oint_C f(z) dz = \operatorname{Res}[f(z), \infty]$

$\oint_C f(z) dz = \sum_{k=1}^n 2\pi i \operatorname{Res}[f(z), z_k]$

(4) $\int_{-\infty}^{+\infty} \frac{x \sin x}{1+x^2} dx$

$\int_{-\infty}^{+\infty} \frac{x e^{ix}}{1+x^2} dx = \int_{-\infty}^{+\infty} \frac{x \cos x}{1+x^2} dx + i \int_{-\infty}^{+\infty} \frac{x \sin x}{1+x^2} dx = i \int_{-\infty}^{+\infty} \frac{x \sin x}{1+x^2} dx$

$\int_{-\infty}^{+\infty} R(x) e^{i\lambda x} dx = 2\pi i \sum \operatorname{Res}[R(z) e^{i\lambda z}, z_k]$

$\therefore$ 原式 $= -i \int_{-\infty}^{+\infty} \frac{x e^{ix}}{1+x^2} dx$

$R(z) = \frac{z}{z^2+1}$ ($m-n=1$) 在上半平面有一级极点 $z=i$, 且在实轴上无孤立奇点.

$\therefore$ 原式 $= -i \cdot 2\pi i \operatorname{Res}[R(z) e^{iz}, i]$

$= +2\pi \lim_{z \rightarrow i} [(z-i) \cdot \frac{z}{(z+i)(z-i)} e^{iz}] = \pi e^{-1}$